NoCap: Fact Checking with AI Milestone 6

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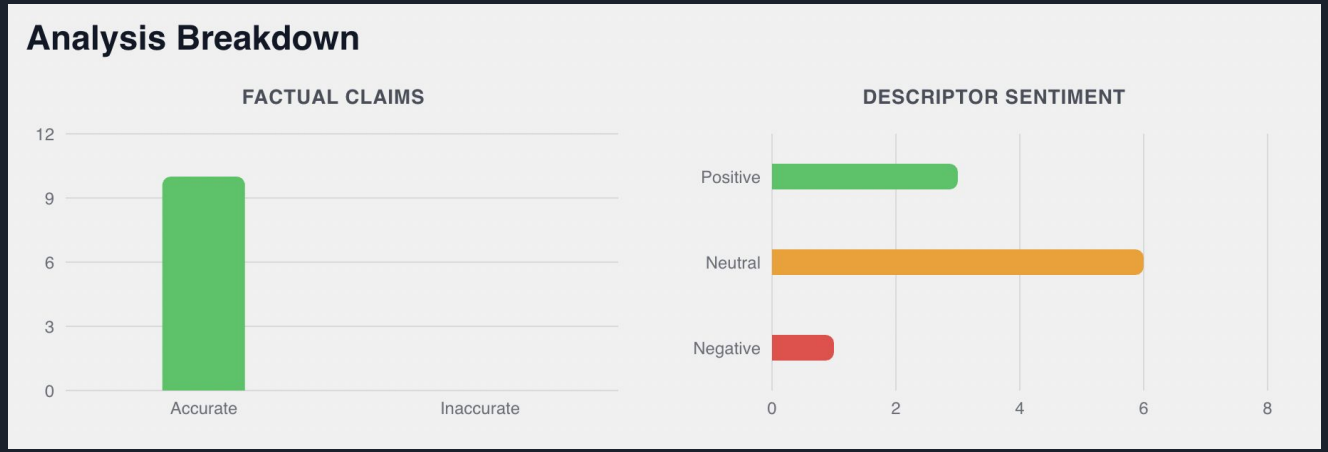
Milestone 5 Matrix

Task	Completion	Thomas	Anthony	Josh	Varun
1. Finalize data graphs/visualizations	100%	70%	0%	0%	30%
2. Cloud deployment	100%	0%	0%	95%	5%
3. Test/demo of the entire system	100%	25%	25%	25%	25%
4. Conduct evaluation and analyze results	100%	20%	40%	20%	20%
5. Create user/developer manual	100%	0%	70%	0%	30%
6. Create demo video	100%	0%	0%	0%	100%

Finalize data graphs/visualizations



Factual Claims Graph:

- Users can see how many claims were accurate vs inaccurate for more transparency.
- Trying to make the system more trustworthy rather than a “black box AI’.

Descriptor Graph:

Not only judges accuracy but HOW it was written.

- Checks narrative framing (positive/negative tone)
- Helps identify misleading but technically true content.

Cloud deployment

Frontend & Database:

- Hosted on AWS amplify gen 2
- CI/CD pipeline using github main
- Serverless

AWS bedrock:

- Hosted on Render
- Manually redeploy newer github main
- Backend server

Test/demo of the entire system

Ensure all features are tested and work as intended

- Home Paged
  - Creating report from URL input
  - Selecting Home Page example cards
- Report Page
  - Reviewing and understanding report information
  - Up/down voting articles
- Database Page
  - Searching/filtering for specific articles
  - Ensure reports are displayed in proper publisher cards
  - Database report information is grabbed and put on report page
- Chrome Extension
  - Generates report, provides summary, adds to database
  - Link to full site to view whole report information

Conduct evaluations and analyze results

- Speed: Reports are created in ~5 seconds, slightly slower upon first opening site as it connects to backend. Grabbed from database instantly
- Accuracy: Improved prompting by adding the current date to increase accuracy of scores, although since it is generated from an AI model we did not create, simply a good estimate. We can not 100% guarantee its validity.
- Performance: Reliably generates reports from input bar and chrome extension, fetches report information from database and puts it to report consistently.
- User Survey: Users perform tasks and ask them questions about ease of use, helpfulness of features, and more on a 1-5 scale.

Create user/developer manual

- Introduction
- User Manual
  - Navigation/Sidebar
  - Home Page
    - Generating an Article Authenticity Report
    - Example Articles
  - Report Page
    - Article Metadata
    - Authenticity Report Information
    - Like/Dislike Feature
    - Graphical Analysis
  - Database Page
    - Report Cards
    - Publisher Cards
    - Searching/Filtering
  - Chrome Extension
- Developer Manual
  - Frontend
    - Pages
      - App
      - Layout
      - HomePage
      - ReportPage
      - DatabasePage
    - Components
      - Sidebar
      - InputBar
      - SearchBar
      - FilterMenu
      - ReportCard
      - PublisherCard
      - AnalysisCharts
    - AppRoutes
  - Backend
    - Database
    - AWS Amplify
    - AWS AI Nova
    - API
    - AI Prompt
  - System Architecture

Final Poster

NORTHROP GRUMMAN
ENGINEERING & SCIENCE
STUDENT DESIGN SHOWCASE
FLORIDA TECH

NoCap: Article Fact-Checking with AI

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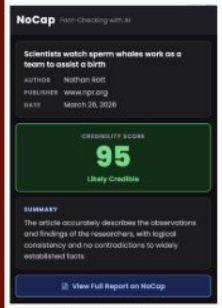
Faculty Advisor(s): Dr. Marius Silaghi, Dr. Philip Chan, Dept. of Electrical Engineering and Computer Science, Florida Institute of Technology

Motivation/Goal

- Fact checking media is important for properly obtaining information
- Create a website that uses AI knowledge to allow users to fact check an article
- Give these articles an authenticity score and report on what is fact or fiction
- Aggregate scores of articles by the same publisher in a database allowing users to view their scores and article authenticity reports

Design/Implementation

- AWS AI Nova Lite: Prompt engineer AWS's Nova Lite model to be used as an article fact checking assistant, only judging factual correctness. Produces a JSON file only returning the authenticity score, summary, and detailed analysis.
- DynamoDB: Used to store the reports so they can be accessed again by anyone from either the database page or inputting the article from the home page.
- AWS Amplify: Used to create a serverless backend to host dynamoDB, our database API.



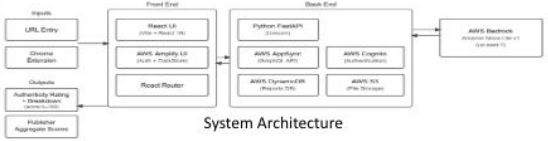
Chrome Extension

NoCap Logo

Home Page

Report Page

Database Page



System Architecture

Approach/Features

- 3 Main Pages
  - Home Page: Users can input the URL of an article they wish to fact check and shows example articles cards found in our database from a range of scores.
  - Report Page: Shows authenticity report, including article metadata, score from 0 to 100, and a summary and detailed analysis of why that score was given.
  - Database Page: Contains all article reports created by users for viewing, sorted by publisher with aggregate score. Filtering options allow for easy searching.
- Chrome Extension: Allows for authenticity report creation simply by fetching current page URL and opening a popup for viewing, same style as report page.
- Graphical Visualization: Two visualizations for reports:
  - 1. Categorizes article claims as Accurate, Inaccurate, or Partially Accurate
  - 2. Tracks adjective and descriptor usage toward people or entities, showing Positive, Negative, or Neutral sentiment

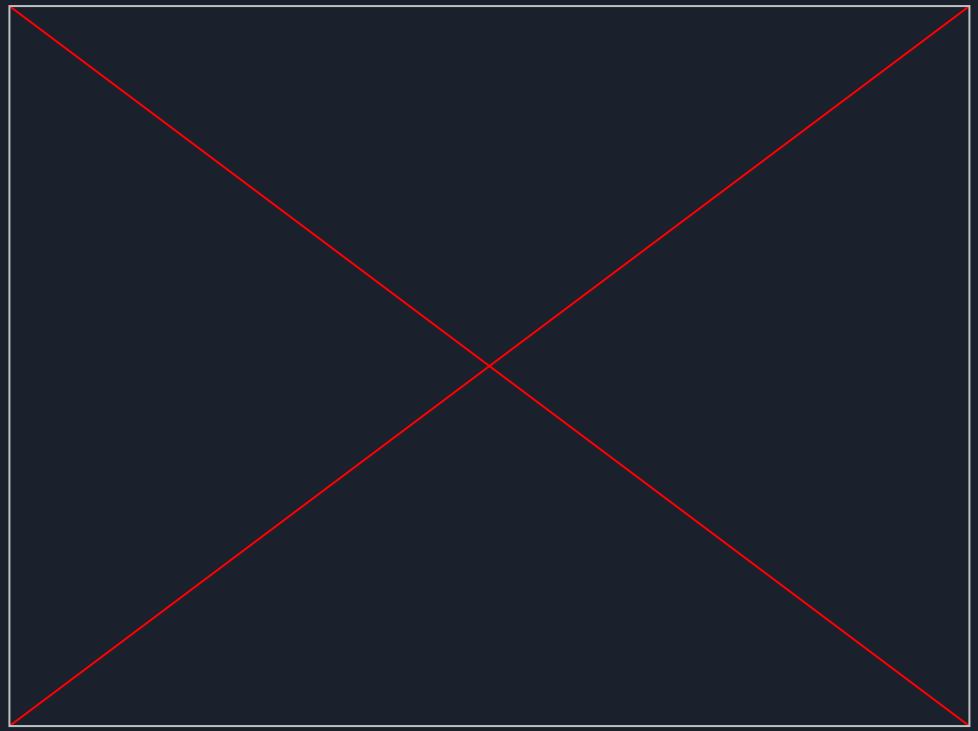
Evaluation Results

- Performance: Achieves report creation and searching/filtering consistently.
- Speed: ~5 seconds to create an article authenticity report and can retrieve reports stored in the database instantly, posting to Report Page.
- Accuracy: Feed the current date to the model and have a mathematical calculation for deriving score so scores and reports are more consistent.
- User Surveys: Task users to use different features and rank their experience on a scale from 1-5.

Limitations/Improvements

- Continuous Prompt Engineering: The model can give a rough estimate on the accuracy of the article, continuous prompt engineering will yield better results.
- Limited Truth Verification: Relies on AI-generated analysis, does not cross-reference claims against authoritative databases or fact-checking organizations.

Create demo video



Lesson Learned

- Creating Real World Product
  - Come up with problem we want to solve (article misinformation)
  - Design a product to help users fight that problem
- Full Stack Application Development
  - New to some team members
  - Connecting front and back end effectively
- Documentation
  - Important in every step (plan, requirements, design, test, milestone)
  - Clear and easy to understand for future referencing
- Collaboration/Splitting Tasks
  - Find each members strengths and weaknesses
  - Clear communication, version control system, teamwork

Thank you
Questions?